

Lab 2: measuring a single pendulum

From high-speed video to a defensible model comparison

Physical model, finite-angle behavior, camera measurement, tracking, and Pre-Lab 2.



What you will do in Lab 2

Release pendulum from rest at three angle ranges.

Record motion with high-speed camera.

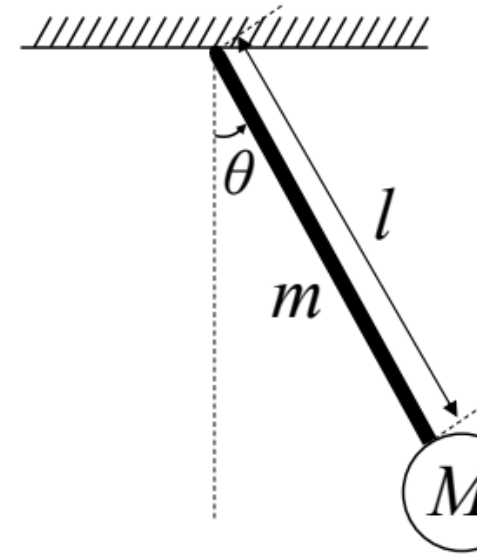
Track bob and convert $x(t)$, $y(t)$ into $\theta(t)$.

Compare period and frequency with model.



Physical system and coordinate

M is the bob and attached hardware mass. m is the uniform bar mass. ℓ is the pivot-to-bob-center distance. $\theta(t)$ is measured from the downward vertical, so $\theta = 0$ at equilibrium. $\omega(t) = d\theta/dt$. Every trial begins from rest, so $\omega(0) = 0$.



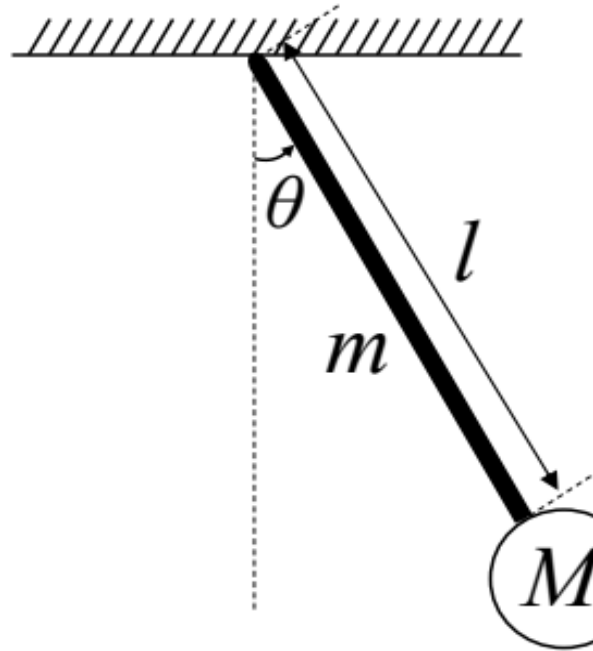
From gravity to the pendulum equation

$$M\ell^2 \, d^2\theta/dt^2 = -Mg\ell \sin \theta$$

$$d^2\theta/dt^2 + (g/\ell) \sin \theta = 0$$

Choose the pivot as the rotation axis. Rotational Newton's law gives $\Sigma\tau = I\alpha$. Gravity applies Mg at the bob, with perpendicular lever arm $\ell \sin \theta$, so $\tau g = -Mg\ell \sin \theta$. For a point mass, $I = M\ell^2$. Combining torque and inertia gives the equation shown above.

Why the real apparatus needs a second model



Moment of inertia about the pivot

$$I_O = M\ell^2 + \frac{1}{3}m\ell^2$$

Gravity torque

$$\tau_g = -(Mg\ell + \frac{1}{2}mg\ell) \sin \theta$$

The bob acts at ℓ . Each piece of the uniform bar lies at a different radius, so the bar contributes to inertia and its weight acts through the bar center of mass at $\ell/2$.

Compound-pendulum equation and natural frequency

$$(M\ell^2 + \frac{1}{3}m\ell^2) \frac{d^2\theta}{dt^2} + (Mg\ell + \frac{1}{2}mg\ell) \sin \theta = 0$$

Small-angle approximation

$$\omega_n^2 = g(M + m/2) / [\ell(M + m/3)]$$

$$f_n = \omega_n / (2\pi) \quad T_0 = 2\pi / \omega_n$$

The coefficient of $d^2\theta/dt^2$ is rotational inertia. The coefficient of $\sin \theta$ is the gravity restoring torque. For small angles in radians, $\sin \theta \approx \theta$. The equation then becomes a linear free-vibration model, and their ratio sets the natural frequency.

Why large initial angle changes period

For small angles in radians, $\sin \theta \approx \theta$ and the response resembles a linear oscillator. At larger angles, $\sin \theta$ is smaller than θ , so gravity provides less restoring torque than the linear model predicts. The pendulum then takes longer to complete a cycle. The three Lab 2 releases test this change directly.



Energy during a swing

$$E = \frac{1}{2}(M\ell^2 + \frac{1}{3}m\ell^2)(d\theta/dt)^2 + (Mg\ell + \frac{1}{2}mg\ell)(1 - \cos \theta)$$

At release, potential energy is largest and angular speed is zero. Near the bottom, potential energy has converted to kinetic energy and angular speed is largest. Pivot friction and air drag remove energy, so amplitude decreases over time. This is visible in $\theta(t)$ and in the angular-velocity plot.

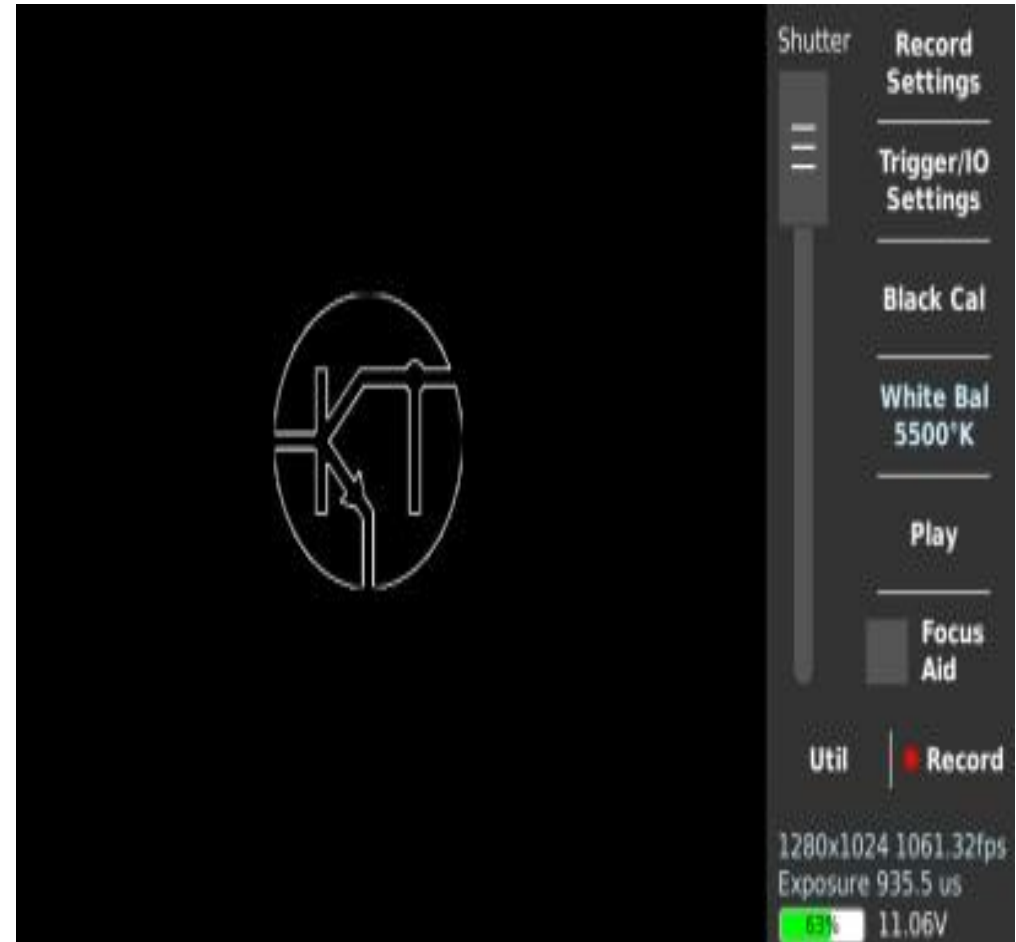
What the camera actually measures

Camera stores frames, not angle directly.

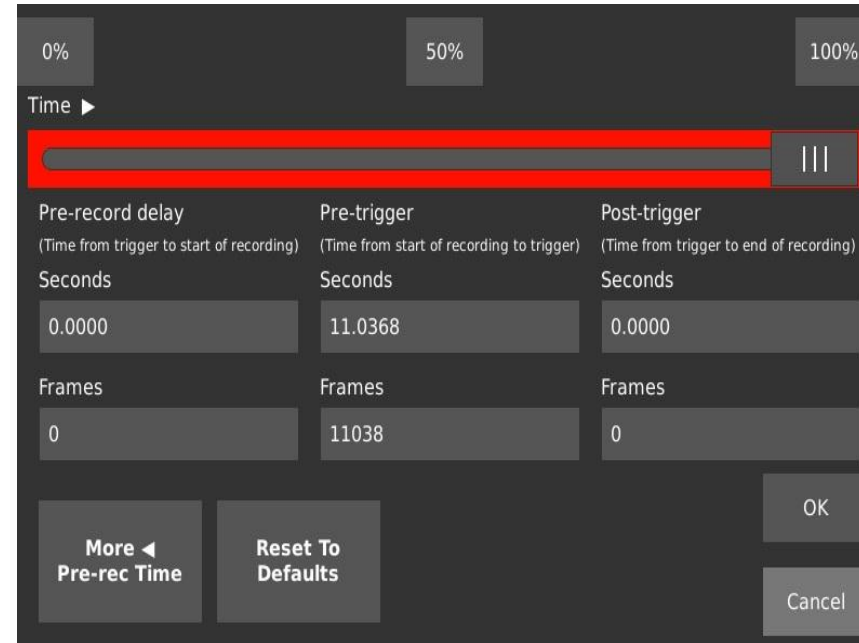
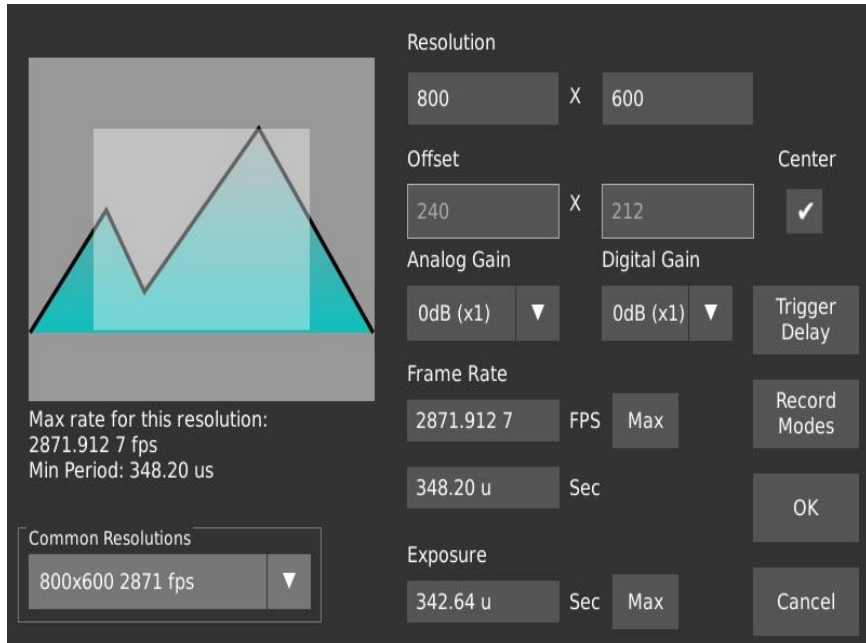
Every frame contains pivot and bob positions.

Useful record needs full motion in frame, sharp contrast, and visible bob center.

Frame rate becomes time spacing of measurement.



Camera settings and useful video



Record actual frame rate.

Use trigger delay that includes release and cycles.

Test framing before experiment.

Smooth video does not prove accurate tracking.

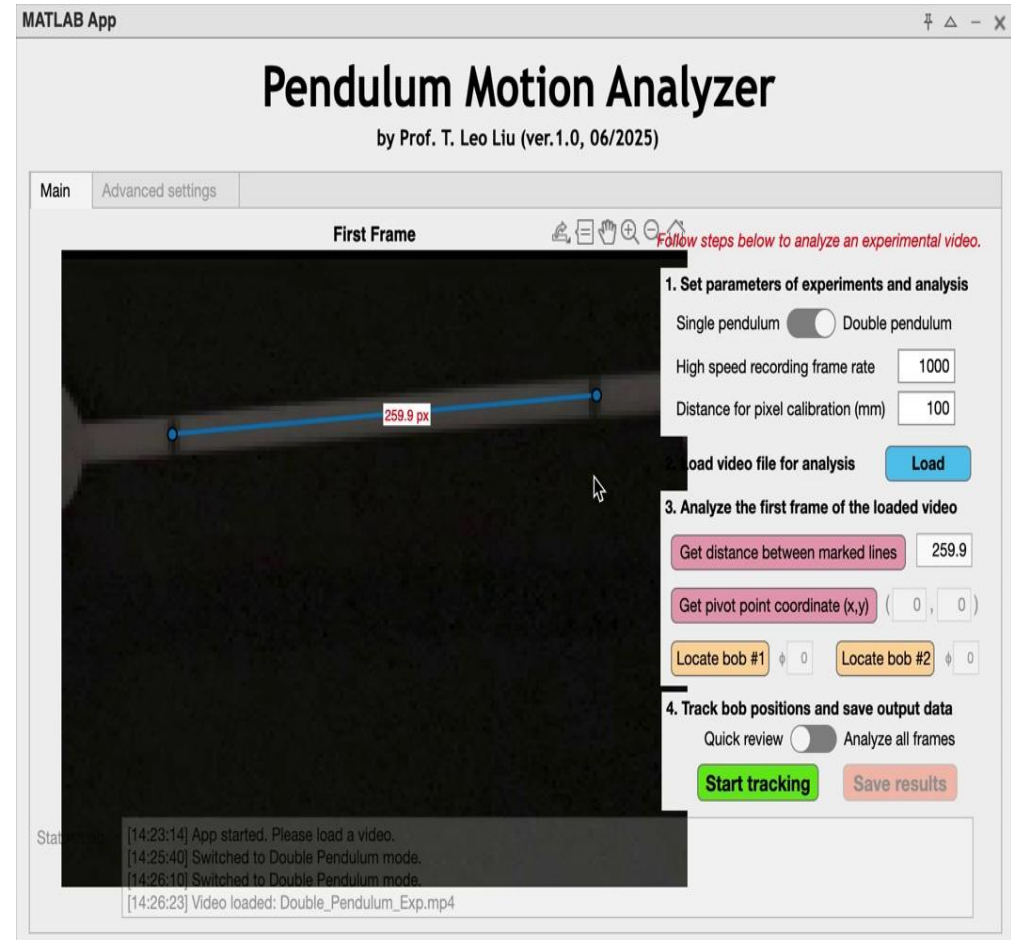
From pixels to coordinates

Calibration marks have known separation, normally 100 mm.

Calibration gives pixel-to-millimeter scale.

Do not move camera or change zoom after calibration.

App saves time and bob coordinates.



From coordinates to a signed angle

$$\theta(t) = \text{atan2}(x_b - x_p, y_b - y_p)$$

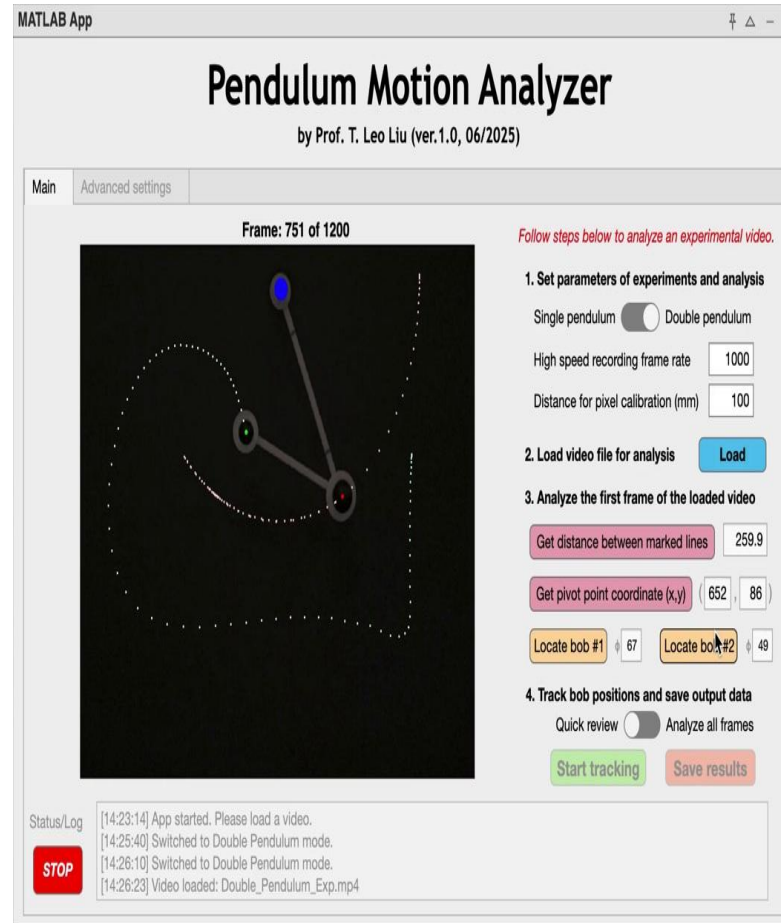
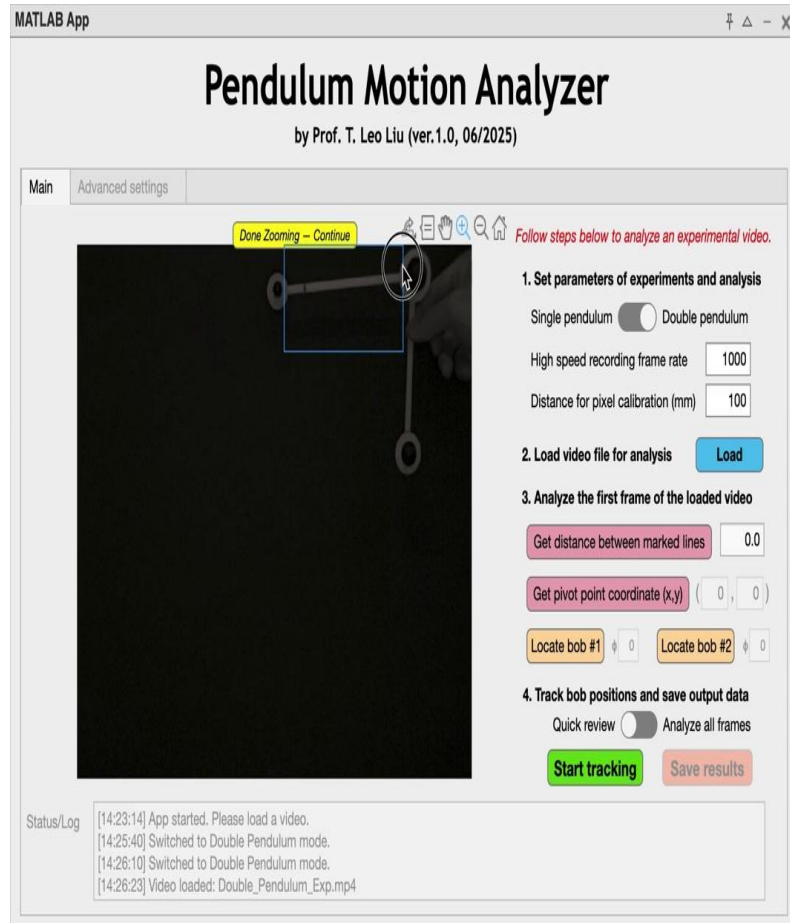
For image coordinates, x points right and y points down.

The formula gives $\theta = 0$ when the bob is directly below the pivot.

Verify zero and sign using one known frame before batch analysis.

Use radians for modeling and derivatives.

Tracking workflow in MATLAB



Select Single pendulum.

Enter actual camera frame rate.

Calibrate, select pivot and bob, then Quick Review.

Run Analyze All Frames only after marker follows bob.

Why frame rate belongs in report

$$f_{\text{Nyquist}} = f_s/2 = 500 \text{ Hz} \quad (f_s = 1000 \text{ frames/s})$$

For $l = 0.82 \text{ m}$, one cycle contains roughly 1800 frames.

The rate greatly exceeds the Nyquist minimum for the pendulum.

Wrong frame rate gives wrong time, frequency, and angular velocity.

Angular velocity and noise

$$\omega(t) = d\theta(t)/dt$$

Differentiation magnifies frame-to-frame error.

Inspect $\theta(t)$ before interpreting $\omega(t)$.

State smoothing method and show it does not hide tracking problems.

Maximum speed occurs near the bottom.

Period from time history and spectrum

Time domain: measure several cycles between same-direction peaks.

$$T = (t_{k+n} - t_k)/N \quad f = 1/T$$

Frequency domain: remove mean, compute FFT, identify dominant peak.

Report record duration T_r and frequency-bin spacing.

$$\Delta f = f_s/N_s = 1/T_r$$

Time-domain period and spectral frequency should agree within resolution.

Lab 2 trial matrix

Trial A: θ_0 below 10 degrees. Closest test of small-angle theory.

Trial B: θ_0 from 10 to 90 degrees. Observe period change.

Trial C: θ_0 from 90 to 180 degrees. Expect strong nonlinearity; keep bob in frame.

All releases start from rest. Record θ_0 from protractor.

Data quality before accepting video

Bob remains in frame.

Bob edge and center are visible without glare.

Pivot and calibration marks are visible.

Release begins from rest.

Quick Review follows bob for nearly all frames.

Notes include file name, angle, frame rate, and calibration distance.

Pre-Lab 2: what you submit

One Word or PDF before laboratory section.

Task 1: compare point-mass and compound-pendulum frequencies.

Task 2: predict behavior before running the supplied simulation.

Task 3: estimate periods and explain finite-angle behavior.

Tasks 4-6: explain coordinates, frame rate, and the laboratory plan.

Evidence for combined Lab 2 and 3 report

Schematic: pivot, l , M , m , θ , and positive direction.

Table: measured parameters and analytical, nonlinear, time-domain, and FFT frequencies.

Plots: $\theta(t)$, $\omega(t)$, and spectra with units.

Discussion: separate model limitations from measurement errors.

In-class prediction

For the same apparatus, which release has the longest period?

- A. 8 degrees
- B. 35 degrees
- C. 105 degrees

Explain using the exact sine restoring torque.

Before leaving laboratory

Save and back up every video and MAT output.

Verify tracking for every trial.

Record initial angle, frame rate, calibration distance, l , M , m , and file names.

Check one $\theta(t)$ plot and one spectrum with the GTA.

Why Lab 2 matters in engineering

Suspended crane loads, robot arms, machinery, and flexible structures can oscillate after a disturbance. Engineers predict natural frequency to avoid resonance and excessive motion. An engineering model predicts a response before operation. Camera data then tests the assumptions. Your report should identify whether a mismatch comes from finite angle, bar mass, damping, or measurement error.